

R823-HG-T003

Moduls de representation: §3

Status: provisional construction test. This is a source-keyed translation experiment, not a native-reviewed standard, not an intelligibility result, and not a completed translation of the R823 volume.

Source boundary. German cumulative authority, lines 21099–21115. Authority SHA-256 begins EE8955BCF7A2639; exact slice SHA-256 begins 73119810BF01CFD. Every translated source segment is aligned in the accompanying clause map. No title, course, or byline metadata is translated in this unit.

§ 3. Moduls de representation

Definition del *modul direct de representation* (d. D. M.).

Un modul direct de representation de $\mathfrak{o}$ relative a $\mathbb{T}$ es cata $\mathfrak{o}, \mathbb{T}$ -bimodul $\mathfrak{M}$. Isto es: $\mathfrak{M}$ es un grup abelian scripte additivmente; $\mathfrak{M}$ porta un action de $\mathfrak{o}$ in le forma cm e un action de $\mathbb{T}$ in le forma $m\tau$. Le duo actiones satisface le lege associativ mixed: per $c \in \mathfrak{o}$, $m \in \mathfrak{M}$ e $\tau \in \mathbb{T}$, se ha $(cm)\tau = c(m\tau)$. In ultra, $\mathfrak{M}$ se pote scriber como suma direct de un numero finit de moduls de $\mathbb{T}$ generate per un sol elemento, con le condition que $x_i\tau = 0$ implica $\tau = 0$:

$$\mathfrak{M} = x_1 \cdot \mathbb{T} + x_2 \cdot \mathbb{T} + \cdots + x_n \cdot \mathbb{T}.$$

Definition del *modul reciproc de representation* (r. D. M.).

Un modul reciproc de representation de $\mathfrak{o}$ relative a $\mathbb{T}^*$ es cata modul $\mathfrak{M}^*$ con un action de $\mathfrak{o}$ in le forma cm e un action de $\mathbb{T}^*$ in le forma τ^*m^* . Le duo actiones satisface le lege de commutation: si $c \in \mathfrak{o}$, $m \in \mathfrak{M}^*$ e $\tau \in \mathbb{T}^*$, in tal caso $c(\tau^*m^*) = \tau^*(cm)$. In ultra, $\mathfrak{M}^*$ se pote scriber como suma direct de un numero finit de moduls de $\mathbb{T}^*$ que le fonte nomina simple, con le condition que $\tau^* \cdot x_i^* = 0$ implica $\tau^* = 0$:

$$\mathfrak{M}^* = \mathbb{T}^* \cdot x_1 + \cdots + \mathbb{T}^* \cdot x_n.$$

Nota editorial de senso

Le parolas de latere non es promovite: le ordine del operandos cm , $m\tau$ e τ^*m^* declara cata action. Le *lege de commutation* afirma que le duo actiones commuta; illo non afirma que $\mathfrak{o}$ o $\mathbb{T}^*$ es un anel commutativ.

In le definition direct, le fonte combina *generate per un sol elemento* con le condition de annulator zero $x_i\tau = 0 \Rightarrow \tau = 0$; le texto non reduce isto al senso moderne plus large de “modul cyclic”. In le definition reciproc, le parola german *einfach* es rendite fonte-a-fronte como *simple*, ma non es promovite como le termino moderne “modul simple o irreducibile” ante revision historic. Le condition de annulator e le formulas controla le uso.

Le fonte declara $\tau \in \mathbb{T}^*$ ma usa τ^* in le formula de commutation; illo usa etiam x_i^* in le condition reciproc ma $x_1, \dots, x_n$ sin asterisco in le formula final. Iste variationes es conservate, non corrigite silentiose. Le signes plus in le duo formulas es etiam conservate, durante que le prosa declara que le decomposition es direct.

Limites del tranche

Le grammatica e le vocabulario es candidatos de construction. Nulle affirmation de comprehension human o de intelligibilitate marginal es facite. Le proxime unitate del fonte comencia al linea 21117, §4, e non es includite hic.